

Module 2 – Data Analysis with Python

Objective

Module 2 introduces students to the fundamentals of Data Analysis using Python. The module focuses on data manipulation, cleaning, visualization, exploratory data analysis (EDA), and deriving business insights from real-world datasets.

Students will gain hands-on experience using industry-standard Python libraries and work on multiple case studies from different business domains.

Learning Outcomes

By the end of this module, students will be able to:

- Import and work with datasets using Python.
 - Perform data cleaning and preprocessing.
 - Explore and analyze datasets using Pandas.
 - Generate descriptive statistics.
 - Create visualizations using Matplotlib and Seaborn.
 - Identify trends, patterns, and relationships in data.
 - Derive meaningful business insights from datasets.
 - Present findings in a professional manner.
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1. Module 2 Notebook – Data Analysis with Python

Notebook Link

https://colab.research.google.com/drive/1FZvww2e66gBFosRQSRAmXgKF_Mdj6YaR

Description

This notebook introduces students to Data Analysis concepts using Python and covers the following topics:

Python Libraries

- NumPy
- Pandas
- Matplotlib
- Seaborn

Topics Covered

- Importing Libraries
 - Reading CSV Files
 - Understanding Datasets
 - Data Exploration
 - Data Cleaning
 - Handling Missing Values
 - Data Filtering
 - Sorting Data
 - Grouping and Aggregation
 - Descriptive Statistics
 - Data Visualization
 - Exploratory Data Analysis (EDA)
 - Deriving Business Insights
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2. Datasets Repository

Google Drive Link

<https://drive.google.com/drive/folders/1YWlb-kXXljTUuRQQqurW9cgDj0CuCvHZ?usp=sharing>

Description

This repository contains all datasets required for Module 2 practical exercises and case studies.

Datasets include:

- Student Performance Dataset
- Movie Dataset
- E-Commerce Dataset
- HR Analytics Dataset
- Healthcare Dataset
- Additional Practice Datasets

Students are expected to download and use these datasets while working on notebooks and assignments.

Case Studies

Case studies provide hands-on exposure to real-world business scenarios. Students are expected to complete all case studies independently and derive meaningful business insights.

3. Case Study – Student Performance Analysis

Notebook Link

https://colab.research.google.com/drive/1SEXaj_DX6q5nOB01aPfNLPsZKxJCydOd

Documentation Link

<https://docs.google.com/document/d/1LI4J5mZNMTOScn9dUP4bTadBtyotvP8oxQ8UPcKUqls/edit?tab=t.0#heading=h.8l2p35h3wpze>

Description

Hands-on case study focusing on student performance analysis using Python and Pandas.

Topics Covered

- Dataset Exploration
- Data Cleaning

- Descriptive Statistics
- Grouping and Aggregation
- Data Visualization
- Student Performance Analysis
- Identifying Top and Low Performers
- Business Insights and Recommendations

Skills Developed

- Data Analysis
 - Problem Solving
 - Visualization
 - Reporting and Interpretation
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4. Case Study – Movie Analysis

Notebook Link

https://colab.research.google.com/drive/1dIKdY_nppnPk2foviVoDE_OxH8fD-EPT

Description

Real-world movie dataset analysis to understand audience preferences and industry trends.

Topics Covered

- Ratings Analysis
- Genre Trends
- Revenue Analysis
- Budget Analysis
- Popularity Analysis
- Visualization Techniques

Skills Developed

- Exploratory Data Analysis
- Trend Analysis
- Data Visualization
- Insight Generation

5. Case Study – E-Commerce Sales Performance Analysis

Notebook Link

<https://colab.research.google.com/drive/1Q4yFkG1herCtpv3LNuQIUZ2osCqc71vM>

Documentation Link

<https://docs.google.com/document/d/1p7jKyCpRK5pprvpCPjrrsu3rEKctsJmQMWIWKQQKGO8/edit?tab=t.0#heading=h.6ovefjuch1aj>

Description

Business-oriented case study focusing on sales performance analysis.

Topics Covered

- Sales Analysis
- Customer Behaviour Analysis
- Revenue Trends
- Product Performance Analysis
- Regional Sales Analysis
- Profitability Analysis

Skills Developed

- Business Analytics
- Data Interpretation
- Dashboard Thinking
- Business Decision Making

6. Case Study – HR Analytics

Notebook Link

<https://colab.research.google.com/drive/127uA7sZkiCTUN08wCna96l047Ltq62ge>

Description

HR Analytics project focusing on workforce and employee analysis.

Topics Covered

- Employee Analysis
- Attrition Analysis
- Department-wise Performance
- Salary Analysis
- Employee Satisfaction Analysis
- Workforce Trends

Skills Developed

- HR Data Analytics
 - Workforce Analysis
 - Business Reporting
 - Strategic Decision Making
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7. Case Study – Healthcare Analytics

Notebook Link

<https://colab.research.google.com/drive/1FPG8l1zNoPHQbmNPgZxbH5uy17EluCNC>

Description

Healthcare analytics case study focusing on patient and healthcare data analysis.

Topics Covered

- Patient Data Analysis
- Healthcare Trends
- Disease Analysis

- Demographic Analysis
- Data Visualization
- Healthcare Insights

Skills Developed

- Healthcare Analytics
 - Data Interpretation
 - Trend Analysis
 - Insight Generation
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Assessment and Examination

Students will be assessed through practical examinations and case study-based analytical tasks.

8. Module Examination Notebook

Notebook Link

<https://colab.research.google.com/drive/1pJzhBkwlgvxqXczfQ9K9quazboVSwMF0>

Description

Assessment notebook used for evaluating student understanding of Module 2 concepts.

The examination assesses students on:

- Data Importing
 - Data Cleaning
 - Data Analysis
 - Visualization
 - Business Insights
 - Problem Solving
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9. Examination Pattern Notebook

Notebook Link

https://colab.research.google.com/drive/11jZt_xPpFGbkb9xozyYOucucv5iWQFYX

Description

Provides students with:

- Examination Format
- Question Patterns
- Expected Analytical Tasks
- Marking Expectations
- Sample Questions
- Evaluation Criteria

Students are strongly advised to thoroughly practice this notebook before appearing for the module examination.

Recommended Practice Guidelines

Students are expected to:

- Complete all notebooks and case studies.
 - Practice Python coding regularly.
 - Perform independent analysis on additional datasets.
 - Create meaningful visualizations for every dataset.
 - Document observations and business insights.
 - Revise all concepts before examinations.
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Industry Skills Developed Through Module 2

Upon successful completion of this module, students will develop the following industry-relevant skills:

- Data Analysis
- Exploratory Data Analysis (EDA)
- Data Cleaning
- Data Visualization
- Business Analytics
- Statistical Thinking
- Insight Generation
- Reporting and Presentation Skills
- Problem Solving
- Decision Making

These skills form the foundation for careers in:

- Data Analytics
- Business Analytics
- Data Science
- Reporting and Business Intelligence
- Operations Analytics
- HR Analytics
- Marketing Analytics